

SAN FRANCISCO BAY RESERVE, CA, USA

WMO: 998011

Lat: 38.2004N

Lon: 122.0260W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 0.6        | 1.9 | -9.5                          | 1.7 | 14.9 | -6.8 | 2.1 | 13.4 | 10.2                  | 12.3 | 9.0 | 12.8 | 1.1                   | 50   | 0.517 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 9             | 13.8                      | 33.8            | 18.9 | 32.3 | 18.4 | 30.4 | 17.9 | 20.3                | 30.6 | 19.6 | 29.2 | 18.9 | 28.0 | 3.5                     | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.6                            | 11.8 | 23.1 | 15.8 | 11.2 | 22.5 | 15.2 | 10.8 | 21.9 | 58.1          | 30.6 | 55.5 | 29.2 | 53.2 | 27.9 | 23.0              |

Extreme Annual Design Conditions

| Extreme Annual WS |     |                    | Extreme Annual Temperature |           |      |            | n-Year Return Period Values of Extreme Temperature |            |      |            |      |      |      |      |      |  |
|-------------------|-----|--------------------|----------------------------|-----------|------|------------|----------------------------------------------------|------------|------|------------|------|------|------|------|------|--|
| Mean              |     | Standard Deviation |                            | n=5 years |      | n=10 years |                                                    | n=20 years |      | n=50 years |      |      |      |      |      |  |
| Min               | Max | Min                | Max                        | Min       | Max  | Min        | Max                                                | Min        | Max  | Min        | Max  |      |      |      |      |  |
| (a)               | (b) | (c)                |                            | (e)       | (f)  | (g)        | (h)                                                | (i)        | (j)  | (k)        | (l)  | (m)  | (n)  | (o)  | (p)  |  |
| 9.0               | 8.2 | 7.6                | DB                         | -1.6      | 36.0 | 1.8        | 2.6                                                | -2.9       | 37.9 | -4.0       | 39.4 | -5.0 | 40.8 | -6.3 | 42.7 |  |
|                   |     |                    | WB                         | -2.5      | 21.7 | 1.7        | 0.7                                                | -3.7       | 22.2 | -4.7       | 22.6 | -5.6 | 23.0 | -6.8 | 23.5 |  |

Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 15.6   | 9.4   | 10.8  | 12.8  | 14.9  | 16.9  | 19.6  | 20.7  | 20.9  | 21.0  | 17.9  | 12.7  | 8.9   |
|                                                                   | DBStd       | 5.25   | 2.58  | 2.69  | 2.86  | 3.40  | 3.17  | 3.00  | 2.70  | 2.70  | 2.87  | 3.03  | 2.90  | 2.82  |
|                                                                   | HDD10.0     | 135    | 42    | 21    | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 9     | 54    |
|                                                                   | HDD18.3     | 1386   | 278   | 211   | 171   | 113   | 67    | 18    | 7     | 6     | 7     | 45    | 169   | 292   |
|                                                                   | CDD10.0     | 2165   | 22    | 43    | 95    | 149   | 214   | 287   | 332   | 338   | 329   | 245   | 90    | 21    |
|                                                                   | CDD18.3     | 374    | 0     | 0     | 1     | 11    | 23    | 56    | 79    | 86    | 86    | 32    | 1     | 0     |
|                                                                   | CDH23.3     | 3784   | 1     | 4     | 17    | 156   | 281   | 622   | 743   | 759   | 847   | 341   | 14    | 0     |
| Wind                                                              | WSAvg       | 3.6    | 2.2   | 2.8   | 3.1   | 3.9   | 4.7   | 4.9   | 5.5   | 5.0   | 3.9   | 2.9   | 2.3   | 2.3   |
|                                                                   | PrecAvg     | 538    | 115   | 103   | 60    | 35    | 24    | 6     | 0     | 1     | 1     | 28    | 53    | 112   |
|                                                                   | PrecMax     | 874    | 291   | 319   | 244   | 116   | 79    | 30    | 1     | 14    | 5     | 115   | 164   | 280   |
|                                                                   | PrecMin     | 281    | 0     | 0     | 0     | 5     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 9     |
|                                                                   | PrecStd     | 159    | 82    | 71    | 64    | 29    | 23    | 8     | 0     | 3     | 2     | 28    | 43    | 83    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 20.5  | 23.7  | 25.5  | 30.7  | 32.9  | 35.6  | 35.3  | 34.9  | 35.9  | 32.9  | 25.3  | 18.6  |
|                                                                   |             | MCWB   | 10.5  | 13.2  | 15.6  | 17.4  | 18.2  | 18.7  | 20.0  | 20.3  | 17.7  | 16.7  | 15.2  | 10.9  |
|                                                                   | 2%          | DB     | 18.4  | 20.3  | 22.8  | 27.7  | 29.6  | 32.6  | 32.6  | 32.8  | 33.3  | 29.9  | 23.0  | 16.5  |
|                                                                   |             | MCWB   | 10.9  | 12.9  | 14.4  | 16.6  | 16.1  | 18.6  | 19.6  | 19.3  | 18.1  | 16.3  | 14.4  | 11.8  |
|                                                                   | 5%          | DB     | 16.3  | 18.0  | 20.9  | 24.5  | 26.3  | 30.0  | 30.0  | 30.2  | 31.6  | 27.0  | 20.7  | 15.1  |
|                                                                   |             | MCWB   | 10.8  | 11.8  | 13.2  | 15.4  | 15.8  | 18.0  | 19.0  | 18.7  | 17.6  | 15.2  | 12.9  | 11.2  |
|                                                                   | 10%         | DB     | 14.6  | 16.1  | 18.8  | 21.6  | 23.5  | 27.2  | 27.4  | 27.6  | 28.5  | 24.7  | 18.3  | 13.8  |
|                                                                   |             | MCWB   | 10.5  | 11.2  | 12.3  | 14.1  | 15.0  | 17.0  | 18.4  | 18.3  | 17.2  | 15.1  | 12.2  | 10.8  |
|                                                                   | 0.4%        | WB     | 14.2  | 15.7  | 16.7  | 19.1  | 19.2  | 21.0  | 21.6  | 21.5  | 20.0  | 19.2  | 17.0  | 15.4  |
|                                                                   |             | MCDB   | 16.5  | 20.0  | 23.8  | 28.7  | 30.9  | 32.5  | 33.5  | 32.6  | 31.2  | 26.9  | 23.1  | 16.5  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 13.0  | 14.4  | 15.2  | 17.4  | 17.7  | 19.9  | 20.2  | 20.2  | 19.1  | 17.9  | 15.6  | 14.1  |
|                                                                   |             | MCDB   | 15.6  | 17.9  | 21.0  | 25.8  | 27.2  | 30.0  | 30.2  | 30.0  | 29.2  | 25.6  | 20.9  | 14.8  |
|                                                                   | 5%          | WB     | 11.9  | 13.3  | 14.2  | 16.0  | 16.5  | 18.8  | 19.3  | 19.2  | 18.5  | 17.2  | 14.5  | 13.0  |
|                                                                   |             | MCDB   | 14.6  | 16.5  | 18.9  | 23.2  | 24.8  | 28.0  | 28.5  | 28.3  | 28.1  | 24.1  | 18.5  | 13.8  |
|                                                                   | 10%         | WB     | 11.0  | 12.2  | 13.3  | 14.8  | 15.4  | 17.7  | 18.5  | 18.5  | 17.8  | 16.3  | 13.5  | 11.9  |
|                                                                   |             | MCDB   | 13.4  | 15.2  | 17.3  | 20.8  | 22.5  | 25.8  | 27.1  | 26.9  | 26.8  | 22.2  | 16.9  | 12.9  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.8   | 9.7   | 10.3  | 11.4  | 11.4  | 12.8  | 13.2  | 13.5  | 13.8  | 12.4  | 10.6  | 8.6   |
|                                                                   |             | MCDBR  | 13.4  | 13.4  | 13.8  | 15.8  | 15.6  | 17.1  | 16.7  | 16.6  | 18.0  | 16.8  | 14.2  | 10.5  |
|                                                                   |             | MCWBR  | 7.1   | 7.2   | 6.6   | 6.9   | 6.3   | 6.5   | 6.2   | 5.8   | 5.6   | 6.3   | 6.6   | 6.0   |
|                                                                   | 5% WB       | MCDBR  | 10.2  | 10.9  | 11.6  | 14.5  | 13.9  | 15.2  | 15.2  | 14.8  | 15.1  | 13.2  | 11.5  | 6.3   |
|                                                                   |             | MCWBR  | 6.6   | 6.5   | 5.9   | 6.7   | 6.4   | 6.6   | 6.2   | 5.8   | 5.5   | 5.4   | 6.1   | 4.5   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.321 | 0.323 | 0.337 | 0.357 | 0.361 | 0.359 | 0.362 | 0.348 | 0.346 | 0.328 | 0.323 | 0.325 |
|                                                                   | taud        |        | 2.531 | 2.530 | 2.478 | 2.398 | 2.411 | 2.421 | 2.410 | 2.442 | 2.451 | 2.534 | 2.528 | 2.521 |
|                                                                   | Ebn at Noon |        | 863   | 910   | 927   | 921   | 918   | 916   | 911   | 919   | 903   | 891   | 852   | 829   |
|                                                                   | Edh at Noon |        | 77    | 87    | 101   | 116   | 117   | 115   | 116   | 109   | 102   | 85    | 76    | 73    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 2.28  | 3.27  | 4.63  | 6.11  | 7.28  | 8.07  | 7.90  | 7.05  | 5.81  | 4.16  | 2.76  | 2.01  |
|                                                                   | RadStd      |        | 0.37  | 0.42  | 0.46  | 0.51  | 0.44  | 0.33  | 0.23  | 0.23  | 0.20  | 0.27  | 0.19  | 0.32  |

Historical Trends

|      | DBAvg                   | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (48 neighbors) | +0.38   | N/A | -1.12  | N/A     | N/A   | N/A   | N/A         | -117    | +131    | N/A     |